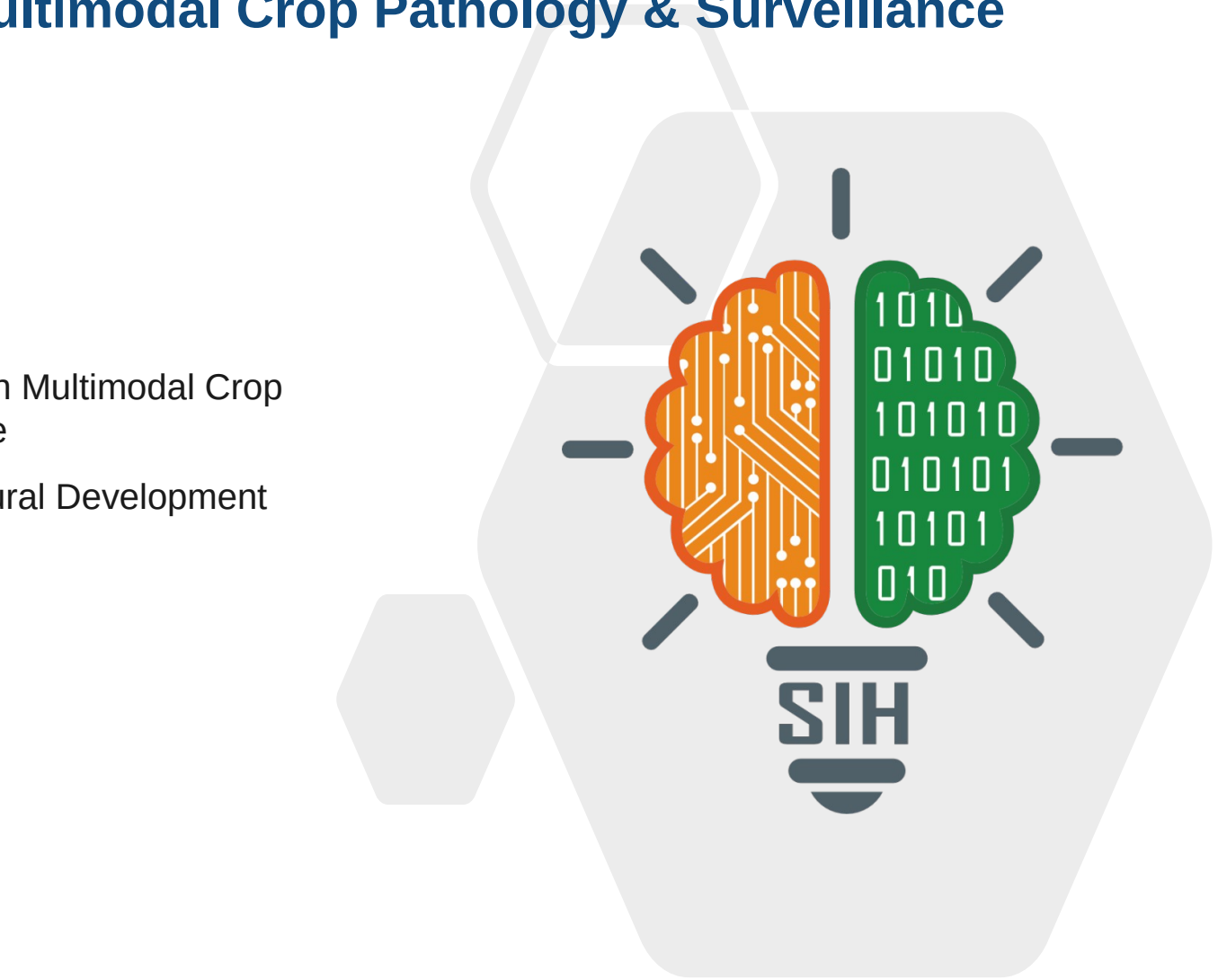
SMART INDIA HACKATHON 2026

AgriTrust AI: Multimodal Crop Pathology & Surveillance

- **Problem Statement ID –** SIH1608
- **Problem Statement Title –** AI-Driven Multimodal Crop Pathology & Trustworthy Surveillance
- **Theme –** Agriculture, FoodTech & Rural Development
- **PS Category –** Software
- **Team ID –** [Your Team ID]
- **Team Name –** binge hack



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IDEA TITLE: AGRITRUST AI

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

- **Detailed explanation:** AgriTrust AI is a multimodal crop pathology engine that integrates ResNet-18 computer vision with real-time soil moisture and 48-hour Open-Meteo weather forecasts to accurately diagnose disease, grade severity via CORN ordinal regression (0-3), and explain detections with Grad-CAM heatmaps.
- **How it addresses the problem:** Standard apps guess blindly on blurry or non-crop photos. AgriTrust implements a Mahalanobis OOD guard that rejects 100% of non-crop objects and abstains below 70% confidence. Furthermore, a cross-crop guard stops misidentifications (e.g. apple leaves scanned under arecanut), while AI chat provides verified, zero-hallucination ICAR/KVK treatment protocols.
- **Innovation and uniqueness:** 1) Dual-modality fusion: 512-dim visual features + 64-dim soil/weather sensor embeddings for outbreak escalation risk. 2) Farmer Bounty Flywheel: "Contribute & Earn" rewards smallholders with micro-payouts for verified field photos through an automated leakage-proof gatekeeper.

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TECHNICAL APPROACH

❖ Technical Approach (Technologies, Methodology & Architecture)

- **Technologies to be used:** PyTorch & Torchvision (ResNet-18 vision models, CORN ordinal loss), SensorEncoder MLP, Open-Meteo REST API (soil moisture & weather), Streamlit Multipage Framework (Python 3.14), WebGL / Pydeck GIS mapping engine, and gTTS for trilingual audio synthesis (English, Kannada, Hindi).
- **Diagnostic Pipeline & Architecture:** Input images first pass through our Ledoit-Wolf Mahalanobis OOD filter to reject invalid objects. Valid inputs are classified across 23 crop pathology classes across Paddy, Arecanut, and Apple, followed by CORN ordinal severity grading and Grad-CAM spatial lesion localization.
- **Multimodal Risk & Surveillance Engine:** Live temperature, humidity, and soil moisture are mapped against plant pathology models (Mills Period for Apple Scab, IRRI Blast indices). Geo-tagged farmer submissions are aggregated to anonymized district centroids on a real-time GIS map, enabling rapid agricultural extension response.

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FEASIBILITY AND VIABILITY

❖ Feasibility and Viability (Implementation, Risks & Mitigations)

- **Feasibility of the idea:** High technical feasibility with lightweight CPU-compatible ResNet-18 checkpoints (~345 MB total). Seamlessly deploys on low-cost cloud servers, local edge devices, or field smartphones with trilingual voice support for low-literacy farmers.
- **Potential challenges and risks:** 1) Non-lab field imagery with complex backgrounds, occlusion, and dust. 2) Accidental submission of incorrect crop categories. 3) Intermittent rural network connectivity.
- **Strategies for overcoming challenges:** 1) OOD Mahalanobis guard calibrated on training residuals prevents false confidences. 2) High-confidence cross-crop verification catches category mismatches. 3) Client-side image preprocessing and cached agronomic rulebooks ensure continuous offline advisory delivery.

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IMPACT AND BENEFITS

❖ Impact and Benefits (Social, Economic & Environmental)

- **Social & Target Audience Impact:** Democratizes agronomic expertise for millions of smallholder farmers across Karnataka and northern apple-growing belts, delivering trusted advisory in Kannada, Hindi, and English with one-click vernacular audio playback.
- **Economic Benefits:** Saves farmers up to 30% in chemical costs by replacing indiscriminate spraying with precision, severity-graded dosage. Additionally, the "Contribute & Earn" bounty wallet directly supplements rural incomes through verified data collection.
- **Environmental & Agronomic Benefits:** Enforces ICAR-approved organic, biological, and soil health management steps before recommending chemical interventions, preventing hazardous pesticide runoff and groundwater contamination.

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RESEARCH AND REFERENCES

❖ Research and References (Literature, Standards & Datasets)

- **Agronomic Protocols & Standards:** ICAR & KVK Package of Practices for Rice, Arecanut, and Horticultural Crops; Mills Period Infection Criteria for Apple Scab (Cornell University Extension); IRRI Climate-Linked Rice Blast Warning Systems.
- **Deep Learning & Methodology References:** He et al. (2016) ResNet Deep Residual Learning; Selvaraju et al. (2017) Grad-CAM Visual Localization; Shi et al. (2023) CORN Ordinal Regression; Lee et al. (2018) Mahalanobis Out-of-Distribution Framework with Ledoit-Wolf Covariance Shrinkage.
- **Datasets & Meteorological Sources:** Paddy Doctor Benchmark (Paddy pathology), PlantVillage & FGVC7 / Cornell University Apple Dataset, Kaggle/Local Arecanut Field Repository, Open-Meteo High-Resolution Global Forecast & Soil Moisture API.